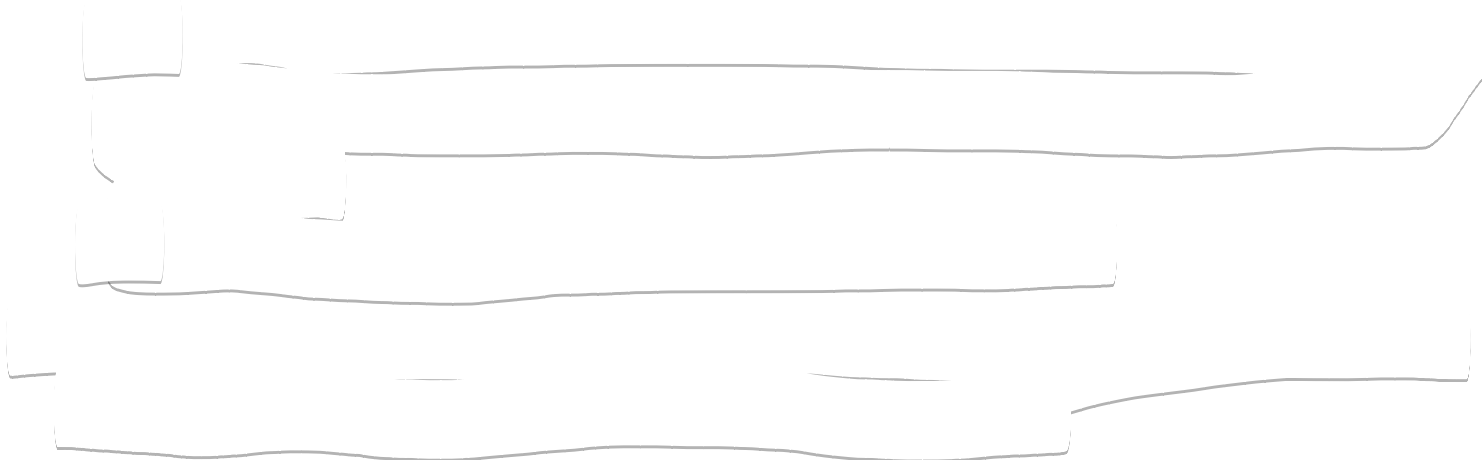


Process Perspectives

- Yet, **additional perspectives** may also be relevant:
 - **Data**: Which data objects are produced or consumed by an activity or checked at a decision point?
 - **Resources**: Which people or systems are involved in the execution of an activity?
 - **Time**: What temporal constraints affect the execution of activities?

Impact on Conformance

- Traces that are conformant according to the control-flow may be non-conforming in according to other perspectives:



- Question of how to balance between the different perspectives

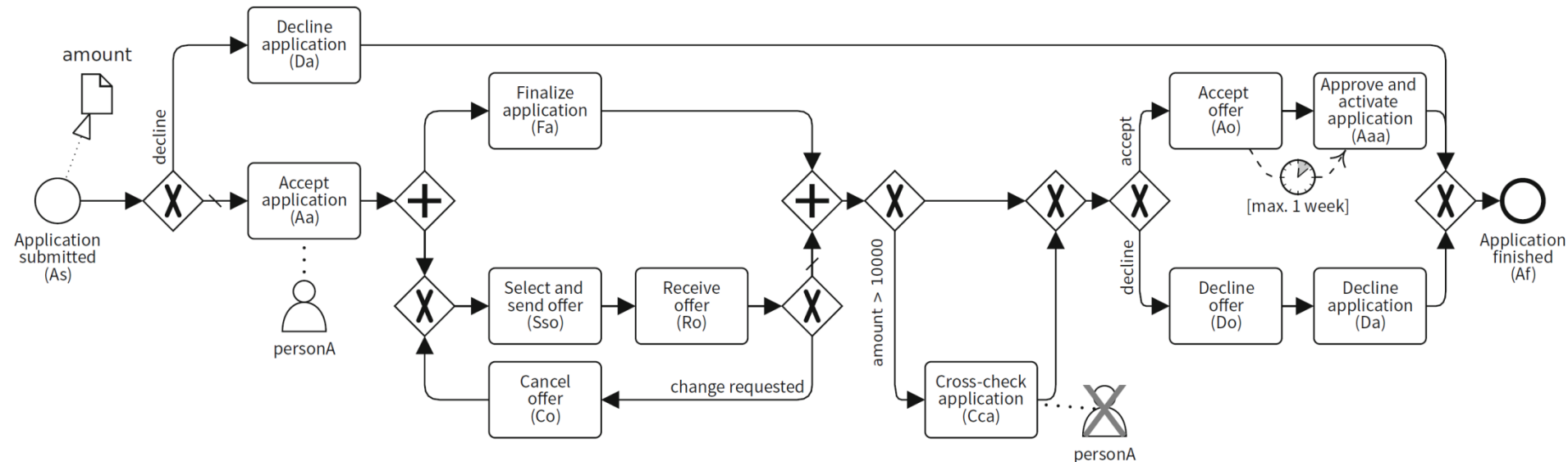
Example Continued

Event	Application	Offer	Activity	Amount	Signed	Resource	Timestamp
e_{13}	A0054		Application submitted	€12,000		<user>	Feb. 1, 12:31
e_{14}	A0054		Accept application			John	Feb. 1, 12:32
e_{22}	A0054		Finalise application			John	Feb. 3, 09:00
e_{37}	A0054	5401	Select and send offer	€11,000		John	Feb. 5, 12:32
e_{42}	A0054	5401	Receive offer		YES	John	Feb. 10, 10:00
e_{54}	A0054	5401	Accept offer			John	Feb. 10, 10:04
e_{64}	A0054		Approve and activate application			John	Feb. 18, 09:05
e_{65}	A0054		Application finished				Feb. 18, 09:06

Using the abbreviations given in the model, the trace when looking only at the control-flow is:

$\langle As, Aa, Fa, Sso, Ro, Ao, Aaa, Af \rangle$

Example Continued



$\langle As, Aa, Fa, Sso, Ro, Ao, Aaa, Af \rangle$

But:

- Amount: 11k => Cross-check application should have been executed
- Time between acceptance of offer and approval and activation of application nearly 8 days => should have been at most a week

Multi-Perspective Alignments

- Idea: Lift alignments from control-flow to multiple process perspectives
 - “Step” in the alignment relates attribute values of a trace of an event log and of an execution sequence of a model to each other
 - Attributes are: Activity, data, resources, time, ...
 - Assigning costs to steps, optimal alignments are defined as before

log trace	execution sequence
<i>As @Feb. 1, 12:31</i> (Res: <user>, Amount: €12,000)	<i>As @Feb. 1, 12:31</i> (Res: <user>, Amount: €12,000)
<i>Aa @Feb. 1, 12:32</i> (Res: John)	<i>Aa @Feb. 1, 12:32</i> (Res: John)
<i>Fa @Feb. 3, 09:00</i> (Res: John)	<i>Fa @Feb. 3, 09:00</i> (Res: John)

Explaining Conformance Issues

- Different perspectives provide different explanations for conformance issues



Example Cost Function

■ Example cost function:

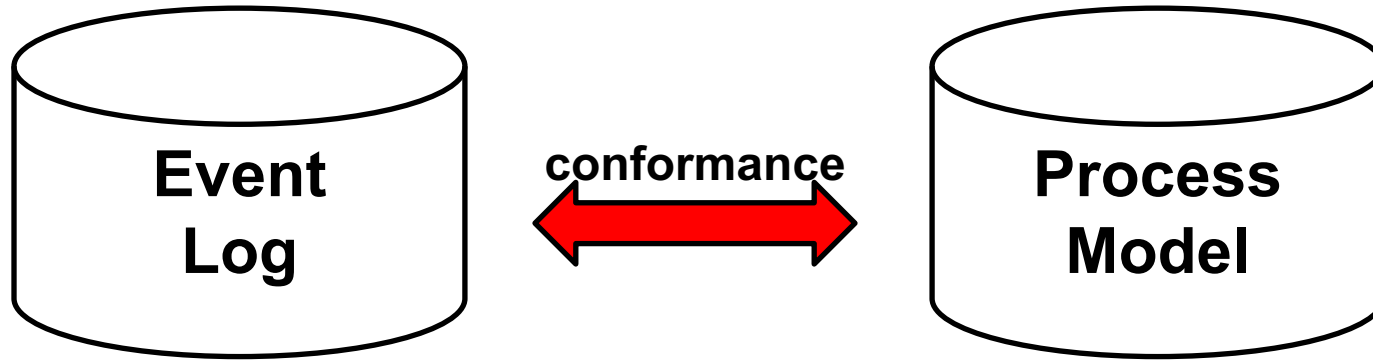
$$c(s_L, s_M) = \begin{cases} 1 & \text{if}(s_L \neq \gg \wedge s_M = \gg) \text{ (move in log)} \\ 1 + |\text{written variables}| & \text{if}(s_L = \gg \wedge s_M \neq \gg) \text{ (move in model)} \\ |\text{unequal written variables}| & \text{if}(s_L \neq \gg \wedge s_M \neq \gg) \text{ (synch. move)} \\ 0 & \text{otherwise} \end{cases}$$

■ Extend standard cost function by penalizing:

- Model moves that write to variables, where the number of written values adds to the cost,
- Synchronous moves that refer to different variables, where the number of different written variables adds to the cost

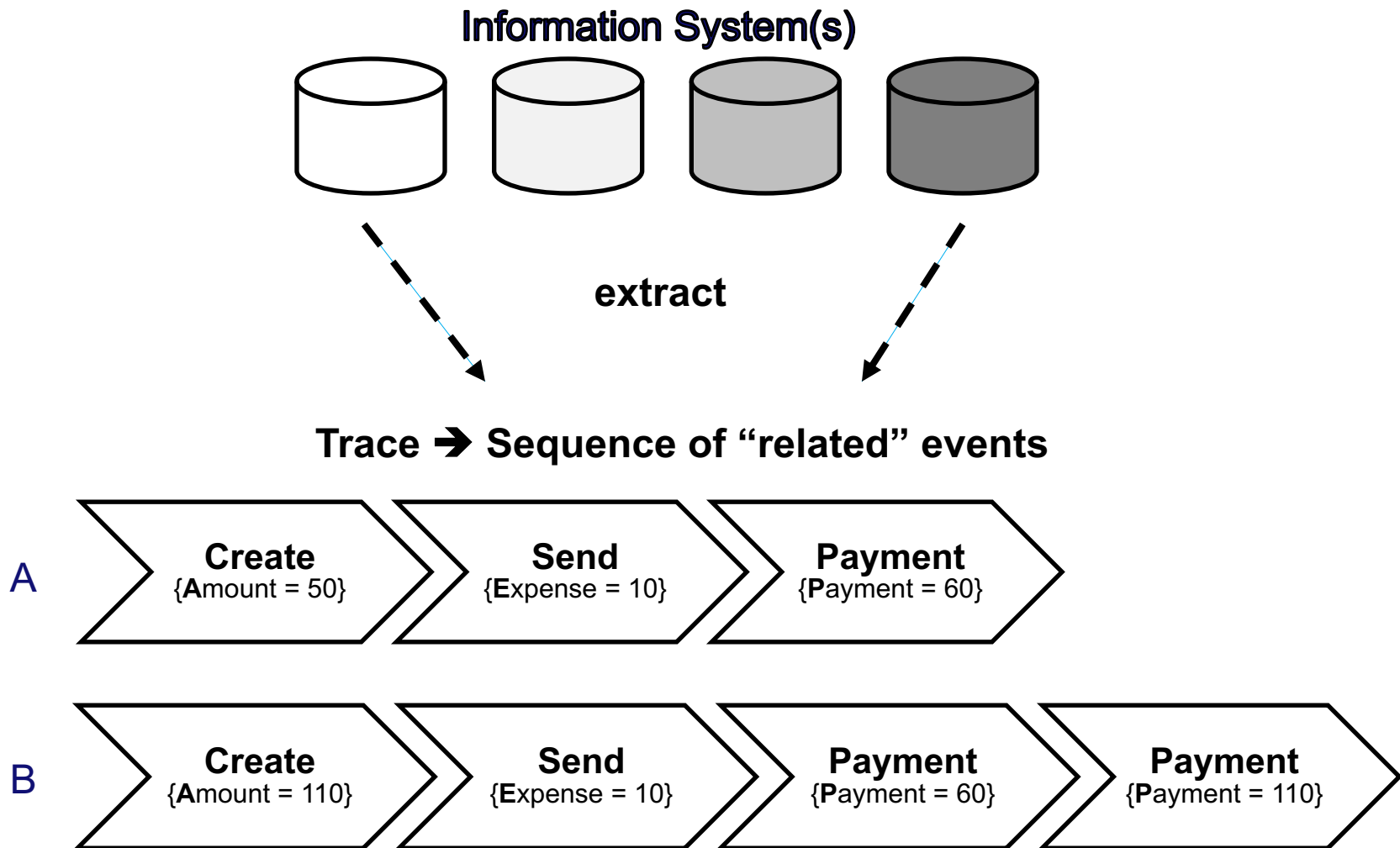
Compute Multi-Perspective Alignments

- **General approach:** Follow the idea of A^* search
 - In each step, pick values for attributes in execution sequence that
 - Are as close as possible to those of the trace
 - Satisfy all constraints imposed by the model
 - However, consider only attribute values that are actually changed by the respective activity
- **But: Computational challenge grows massively**



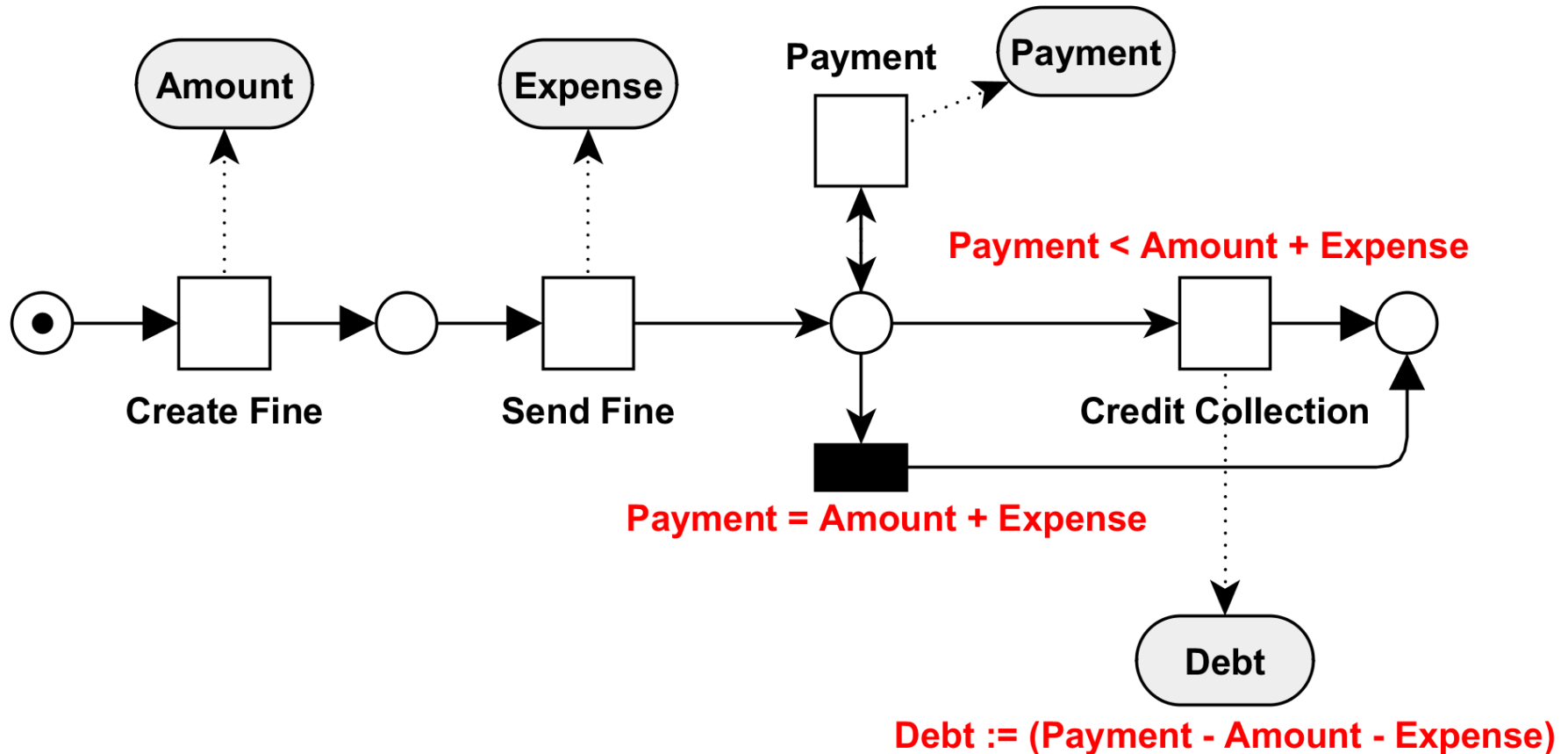
Data-aware Process Conformance

Ingredient A: Event Log



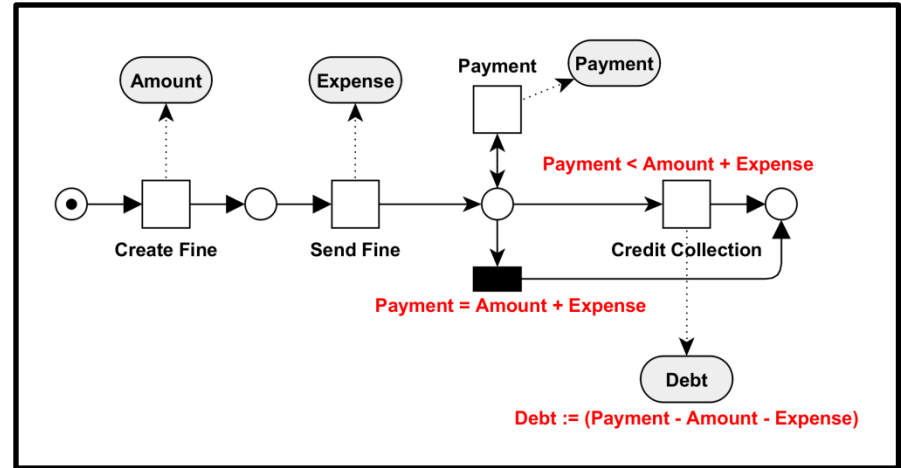
Ingredient B: Data-aware Process Model

Process Model



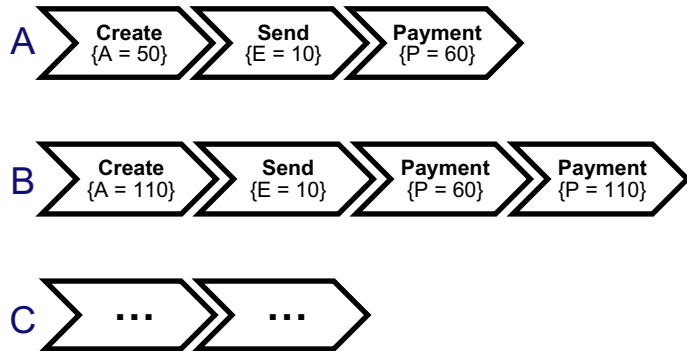
Process Conformance

Process Model

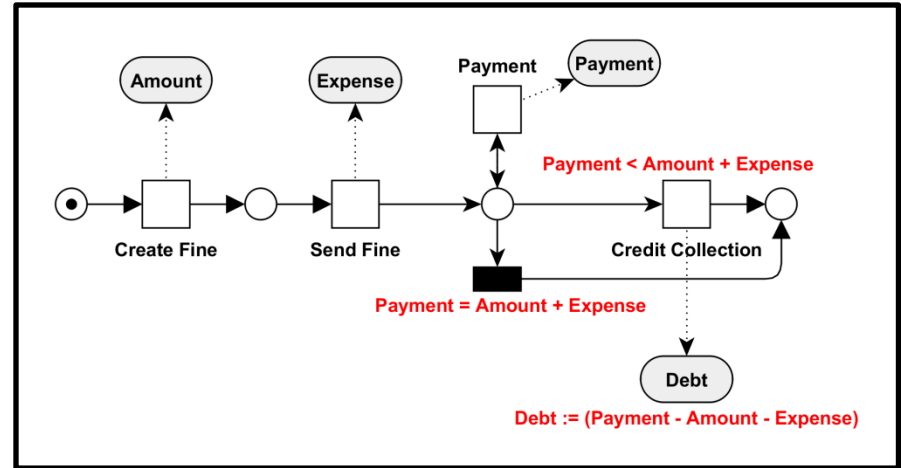


Process Conformance

Event Log

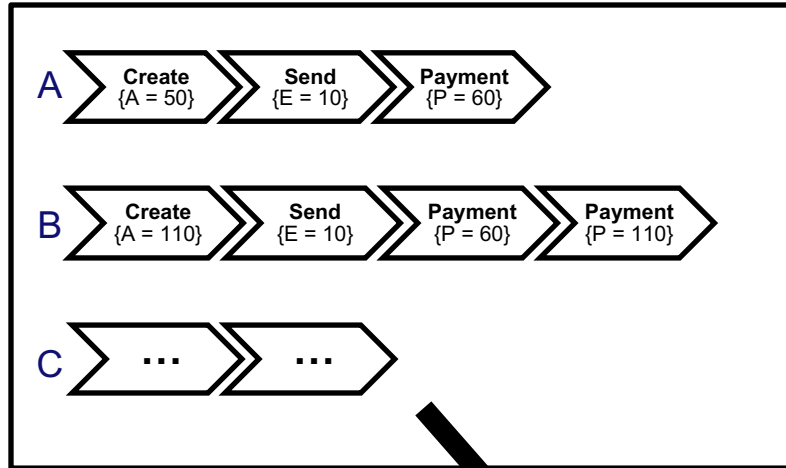


Process Model

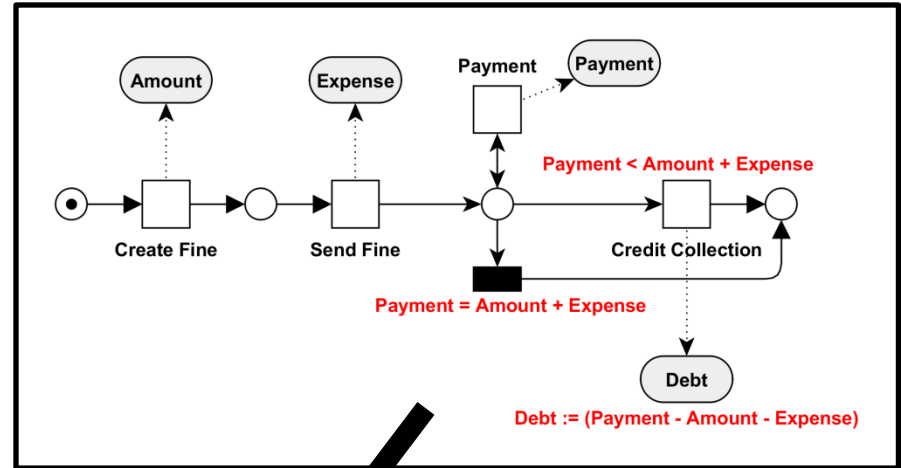


Process Conformance

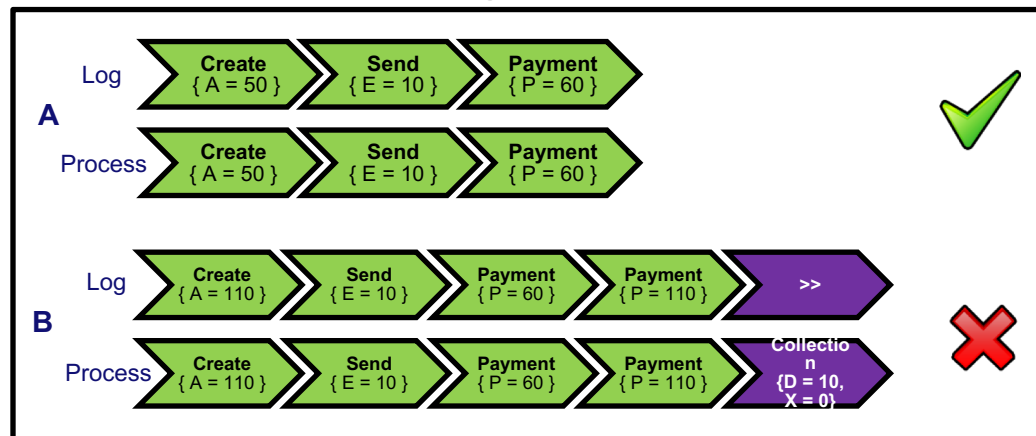
Event Log



Process Model

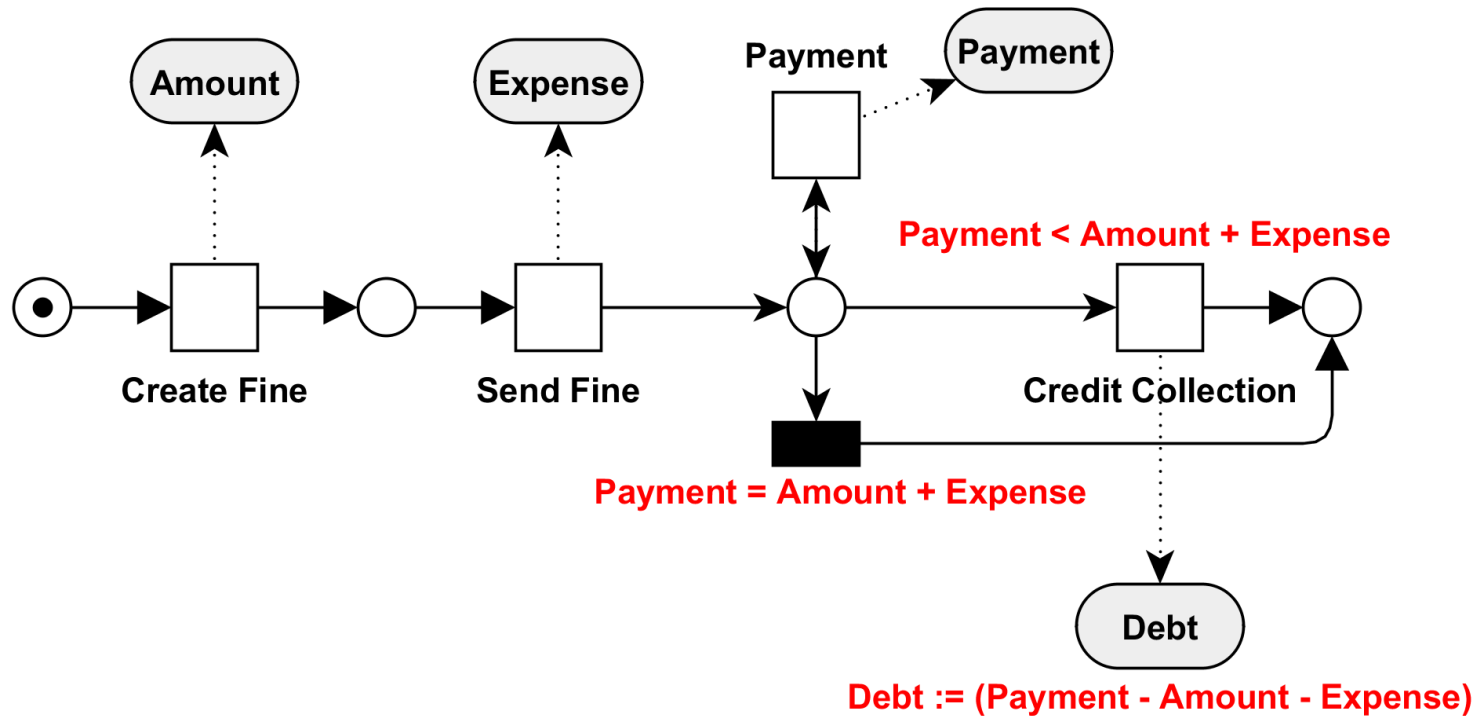


Alignment

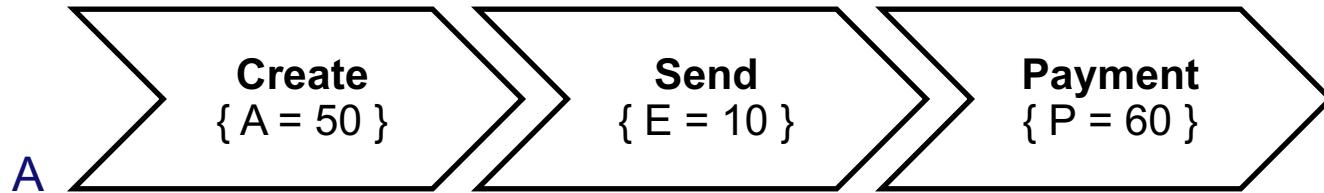


Process Model: Data Petri Nets

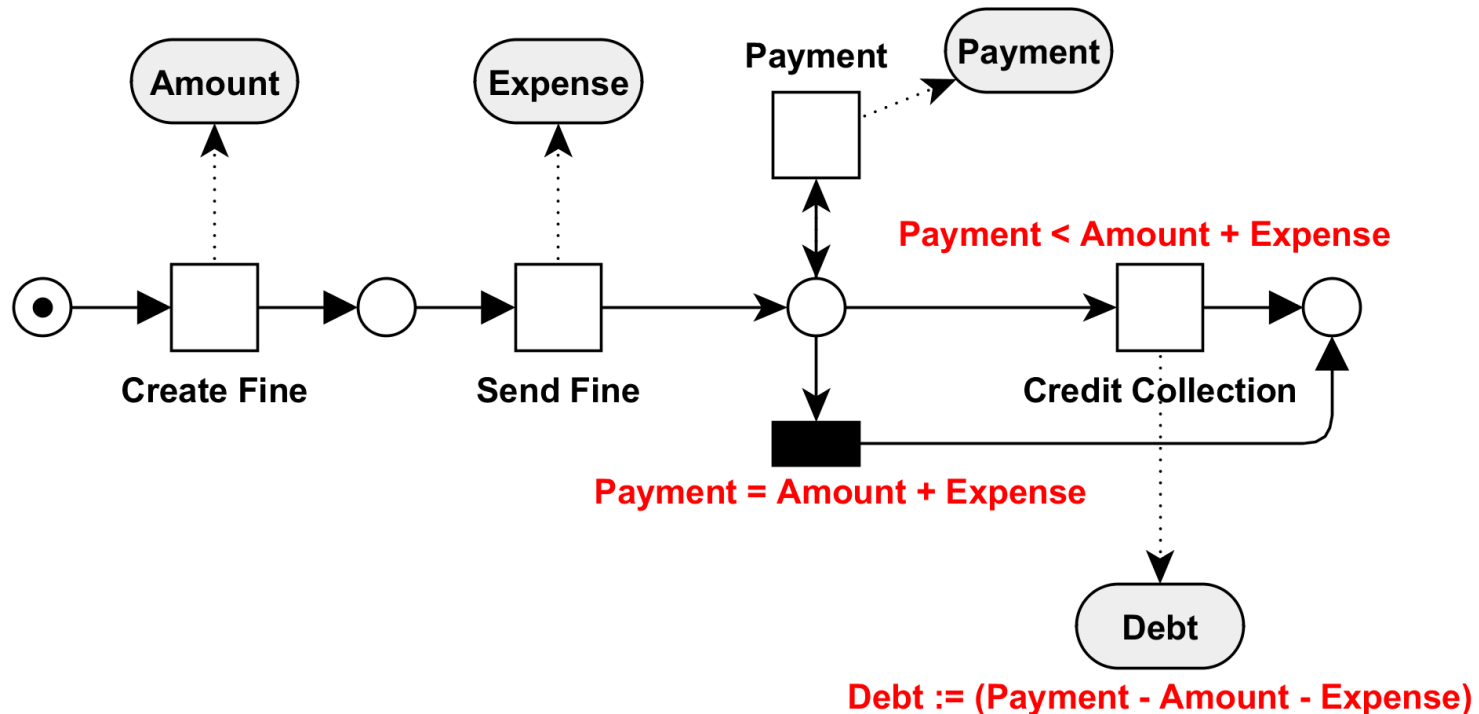
Process Model



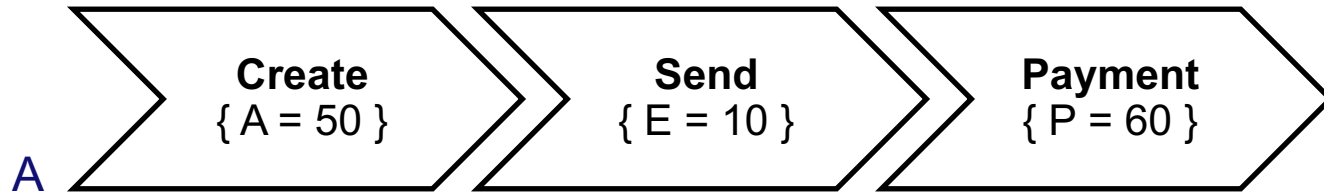
Process Model: Data Petri Nets



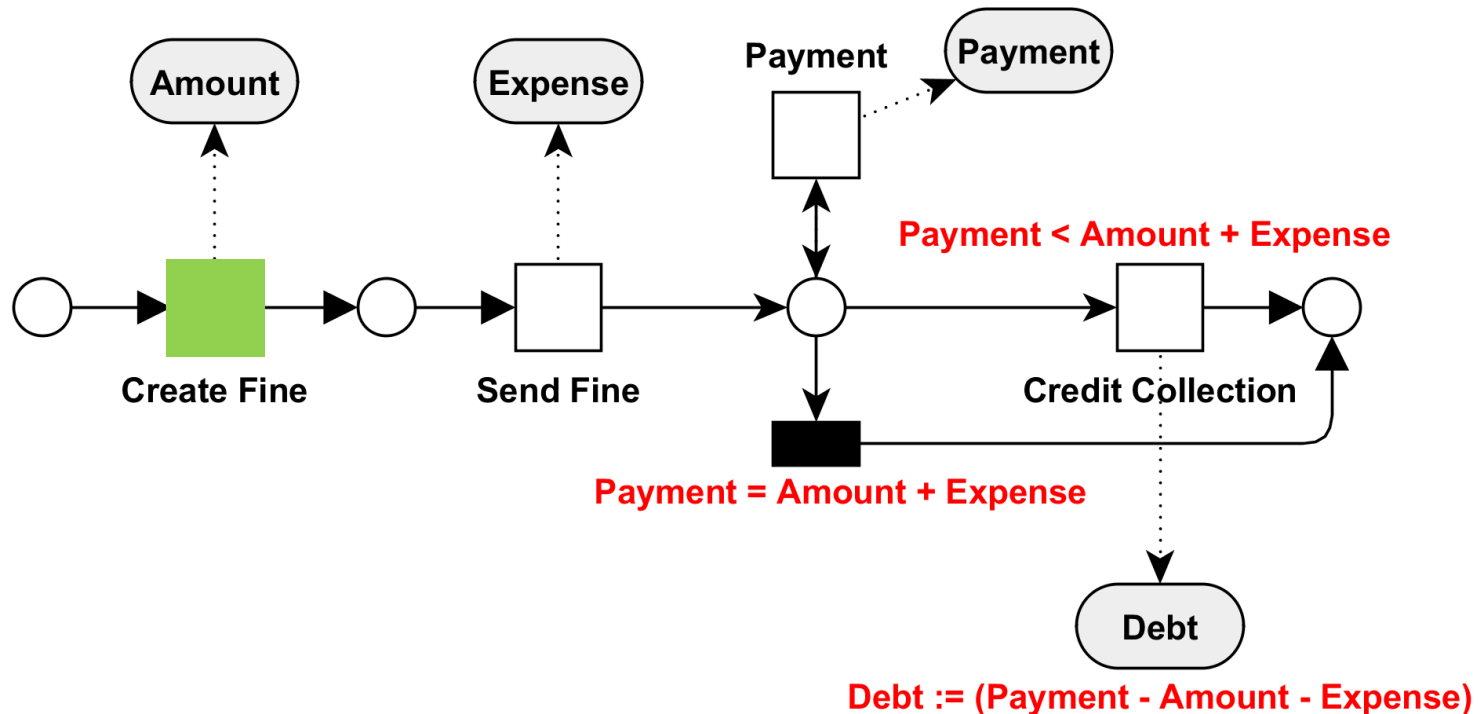
Process Model



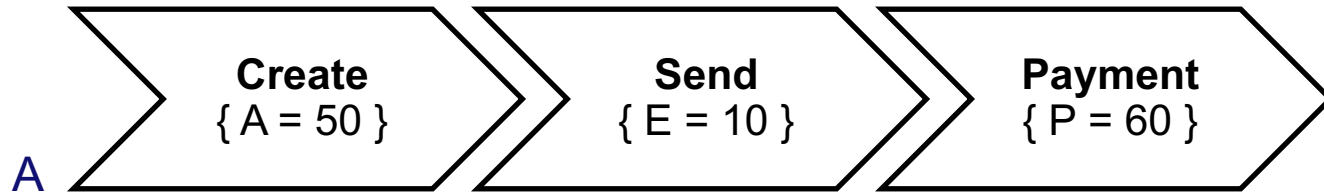
Process Model: Data Petri Nets



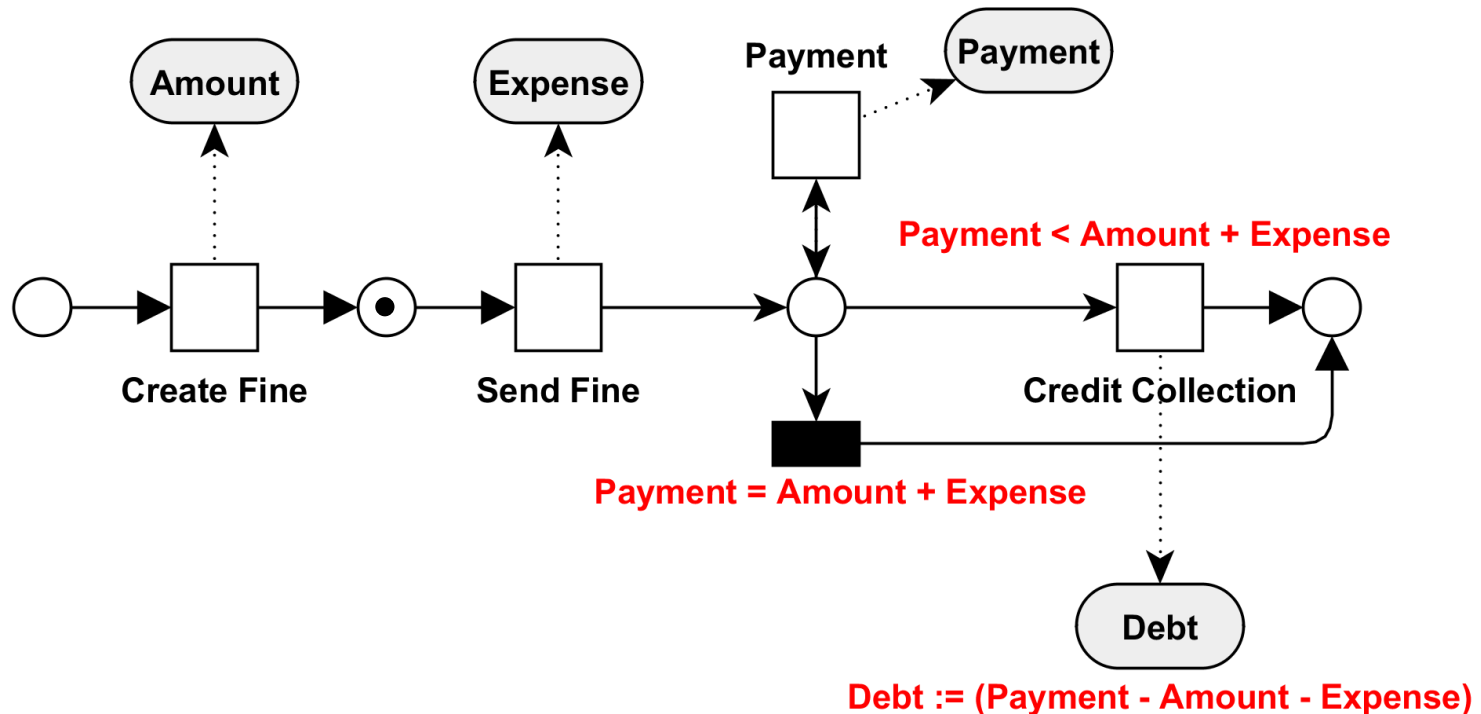
Process Model



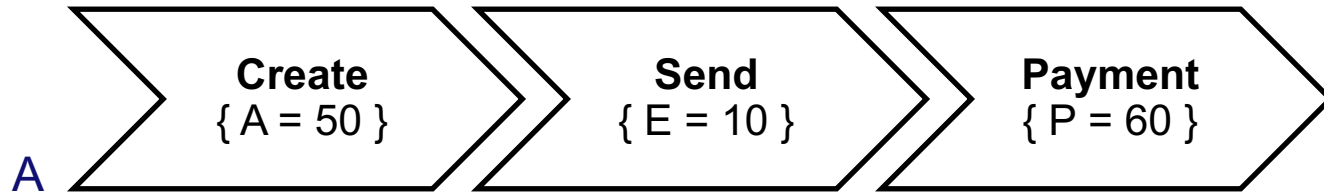
Process Model: Data Petri Nets



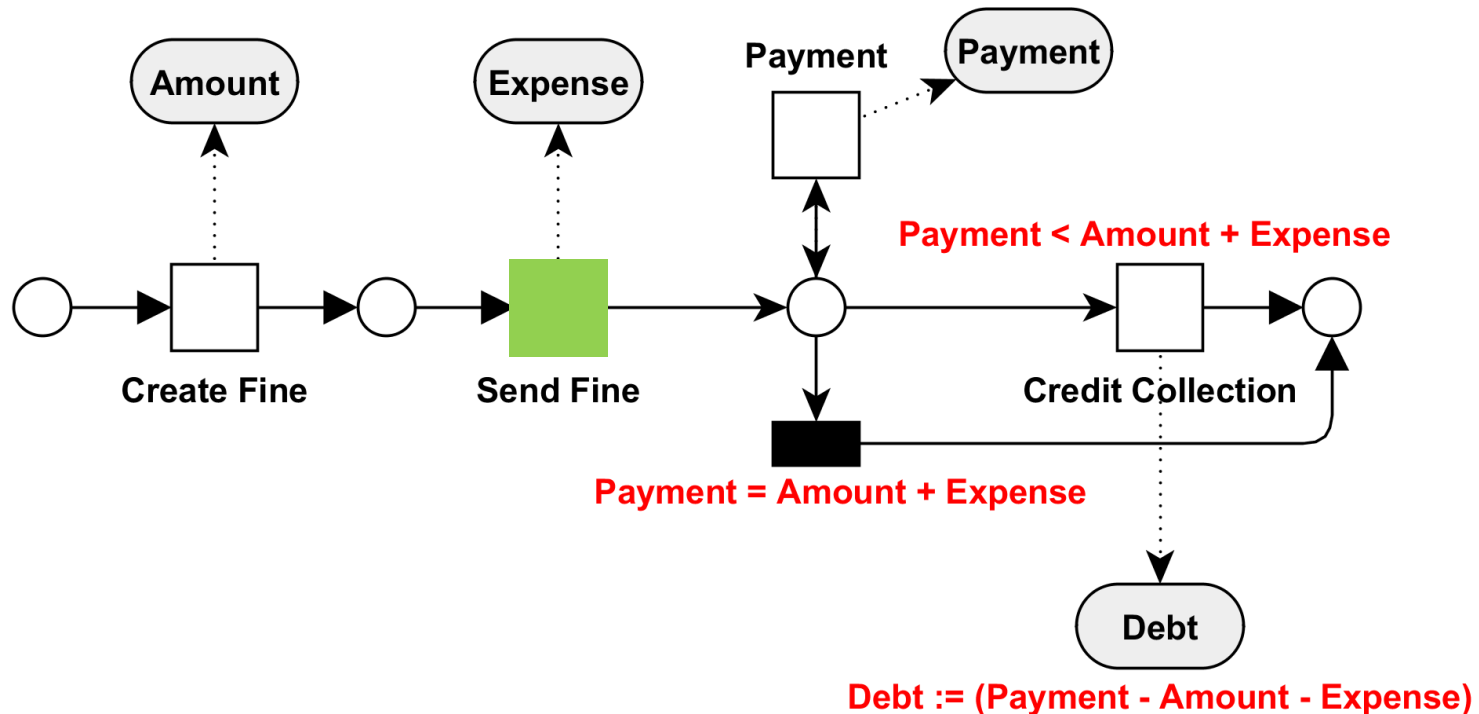
Process Model



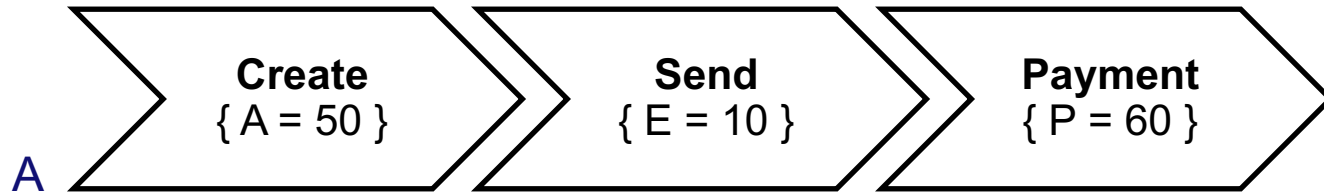
Process Model: Data Petri Nets



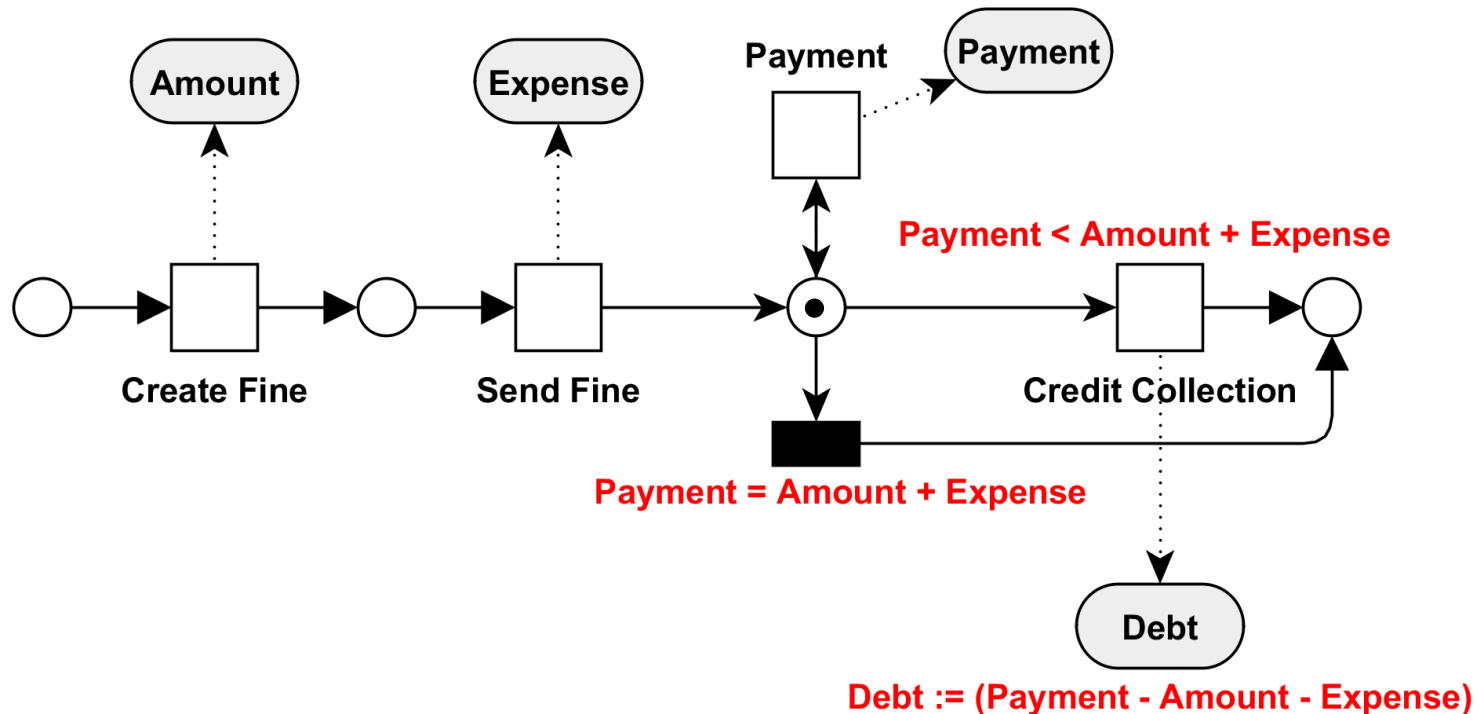
Process Model



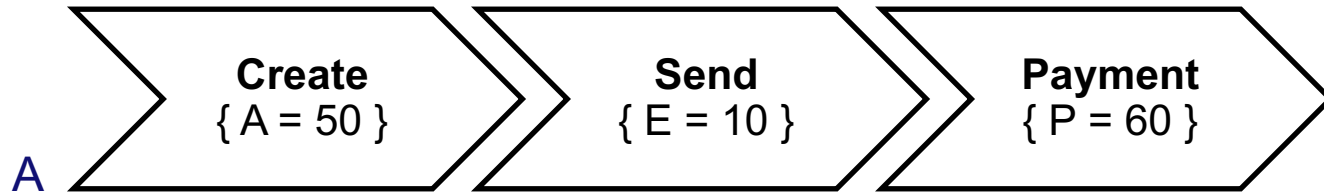
Process Model: Data Petri Nets



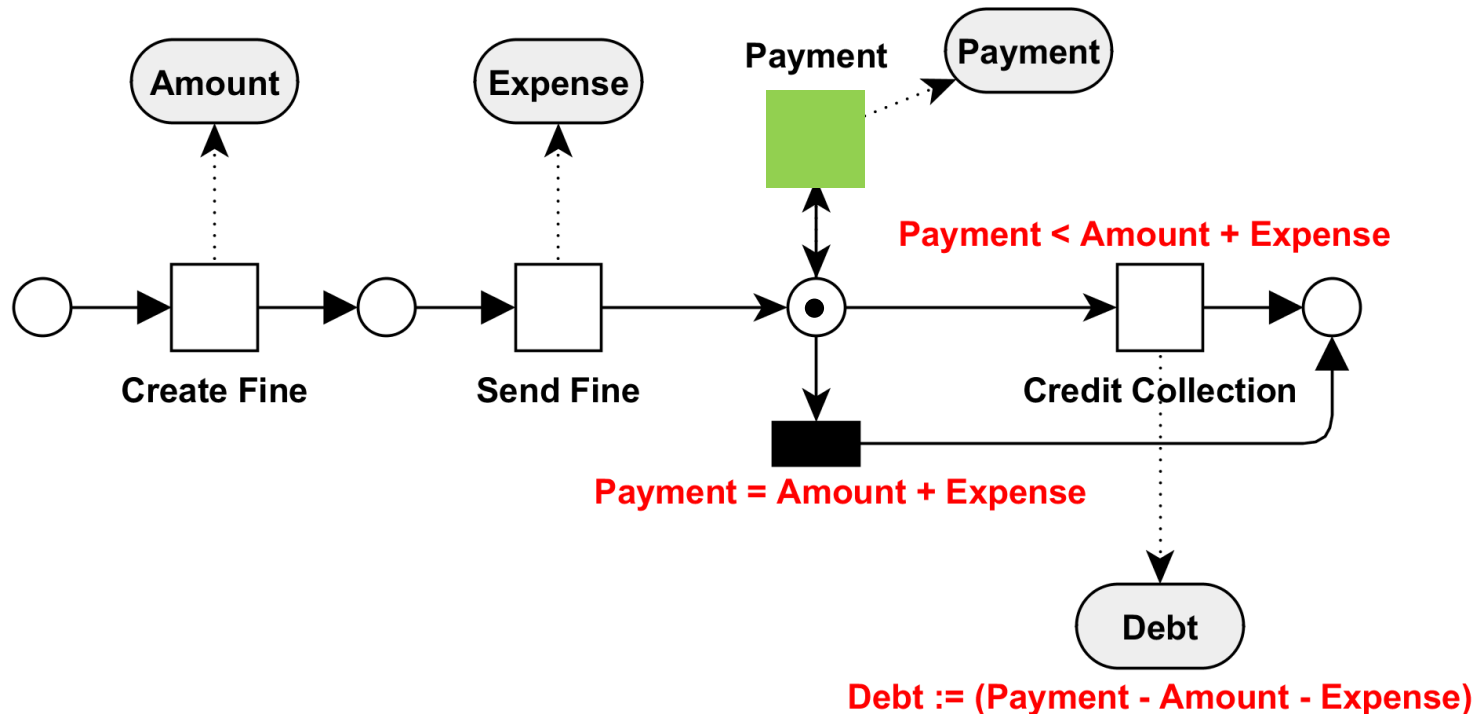
Process Model



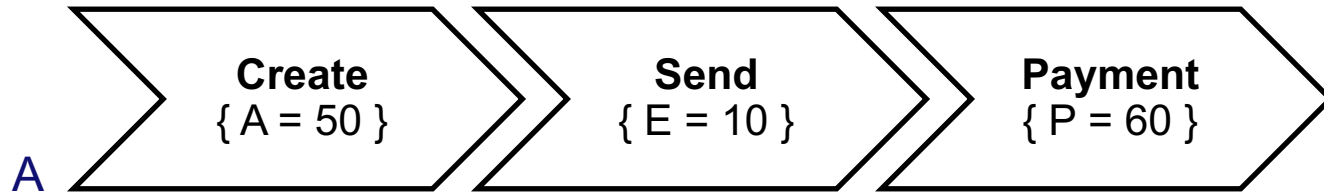
Process Model: Data Petri Nets



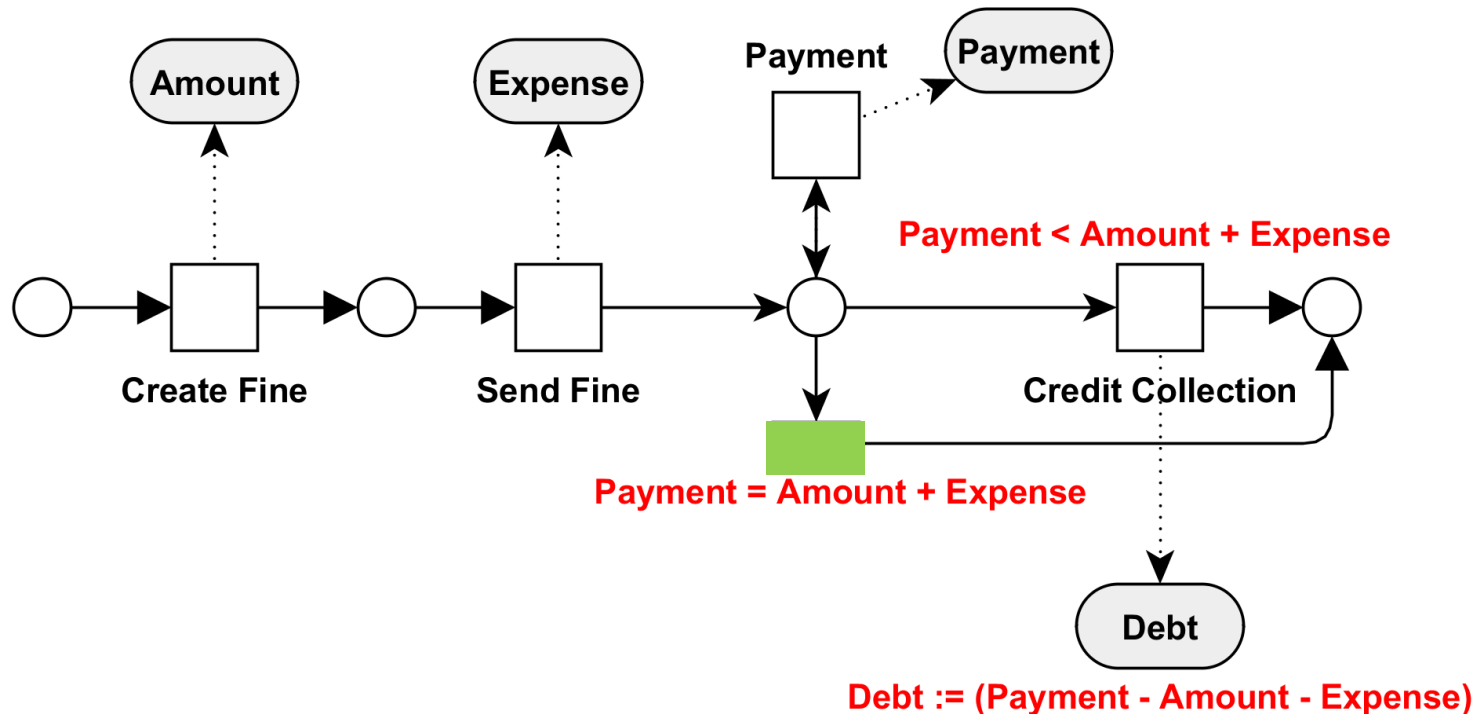
Process Model



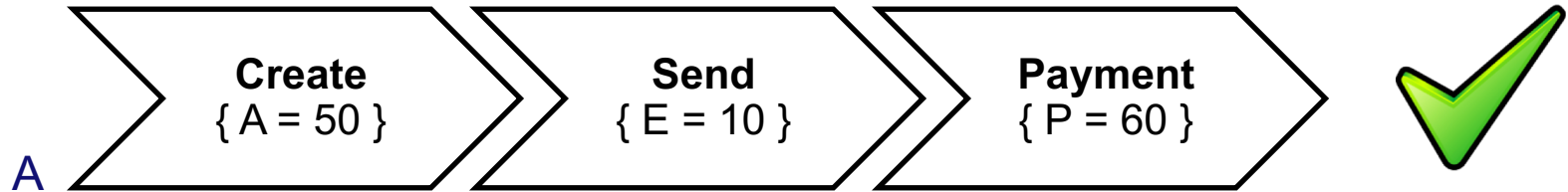
Process Model: Data Petri Nets



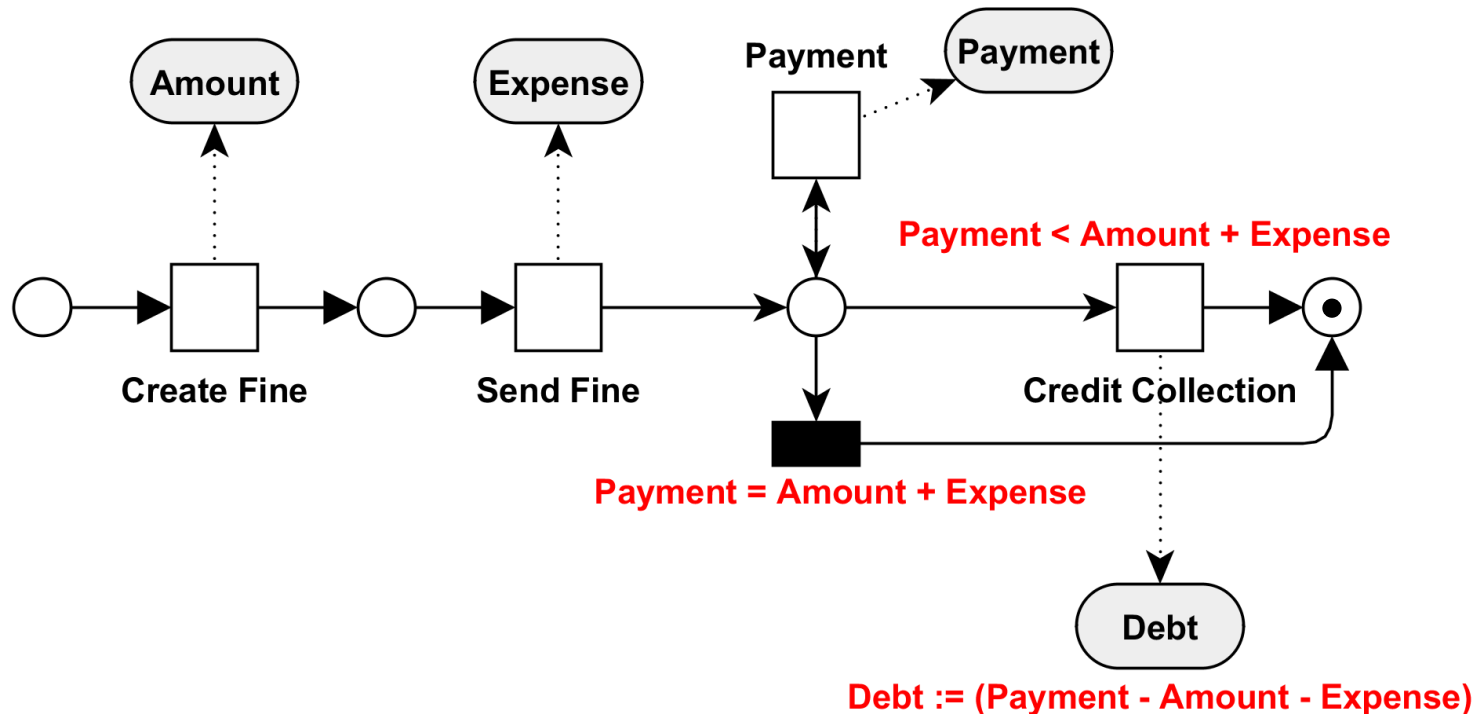
Process Model



Process Model: Data Petri Nets

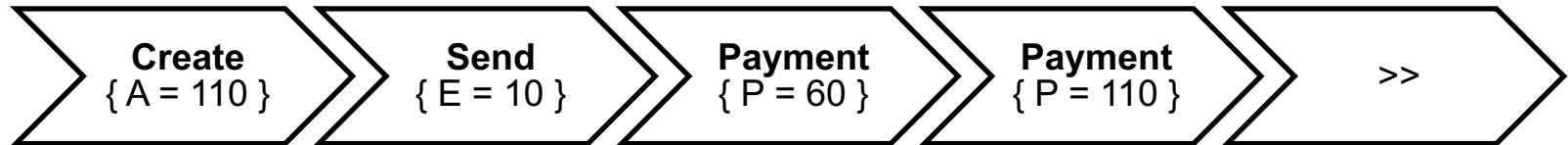


Process Model

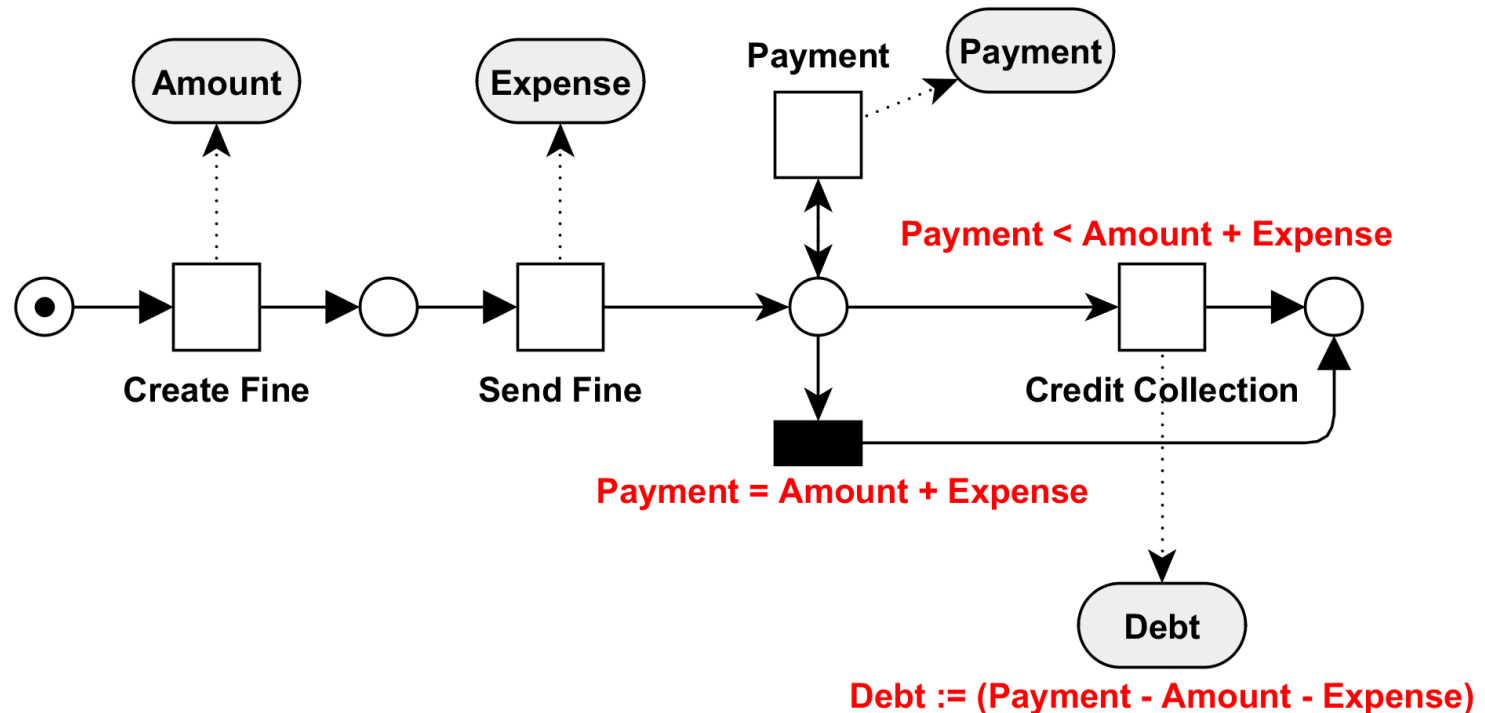
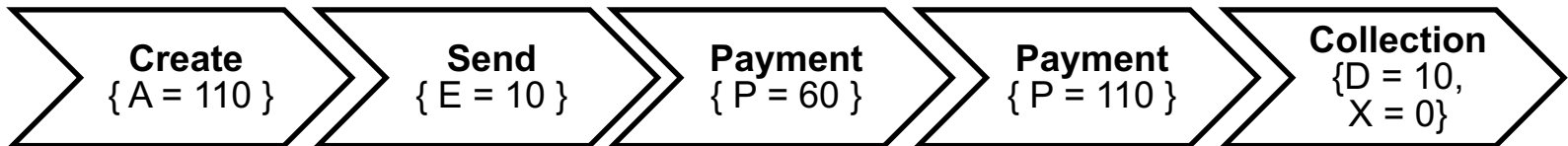


Alignment: Searching for a likely Explanation

Log

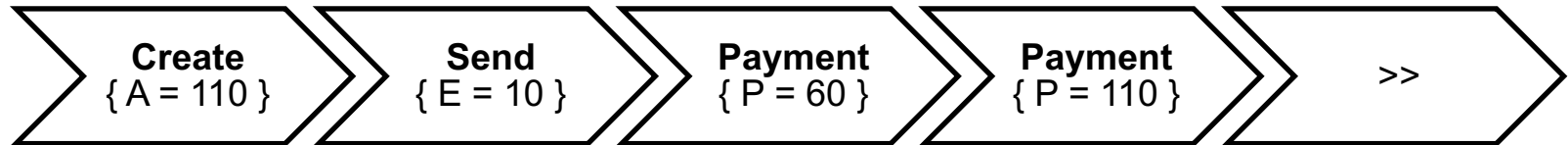


Process

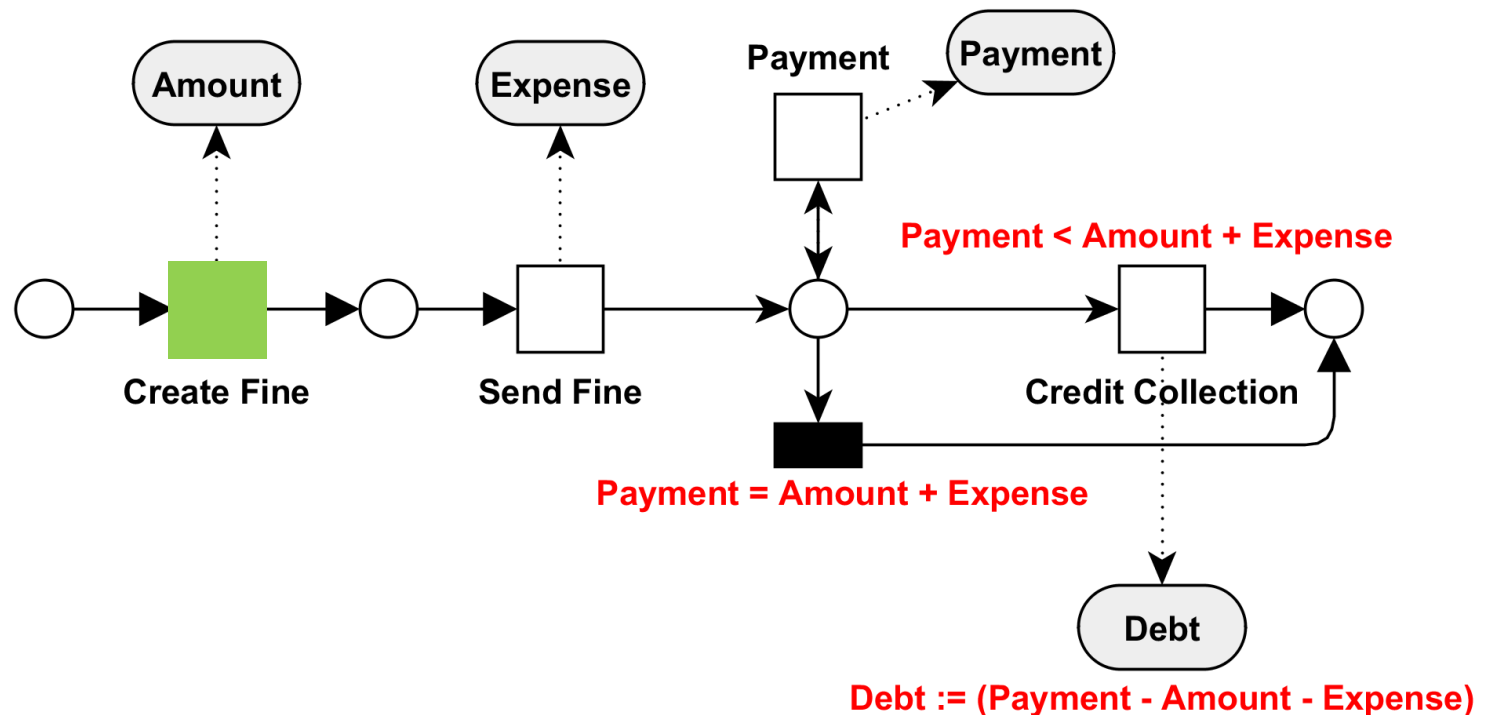
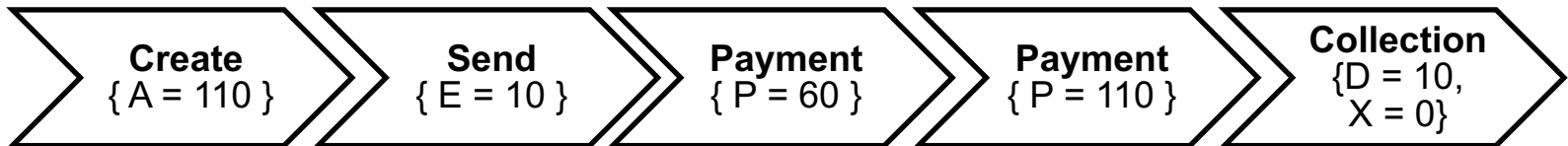


Alignment: Searching for a likely Explanation

Log

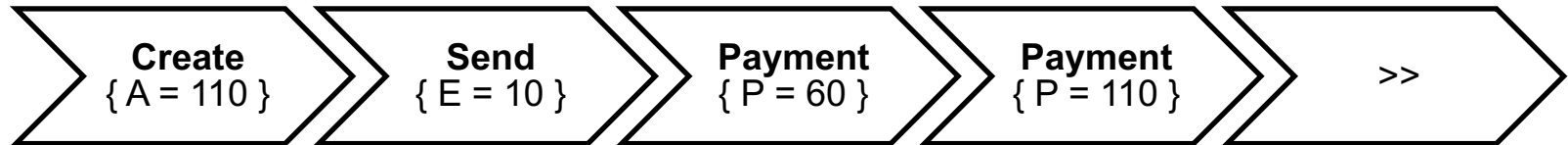


Process

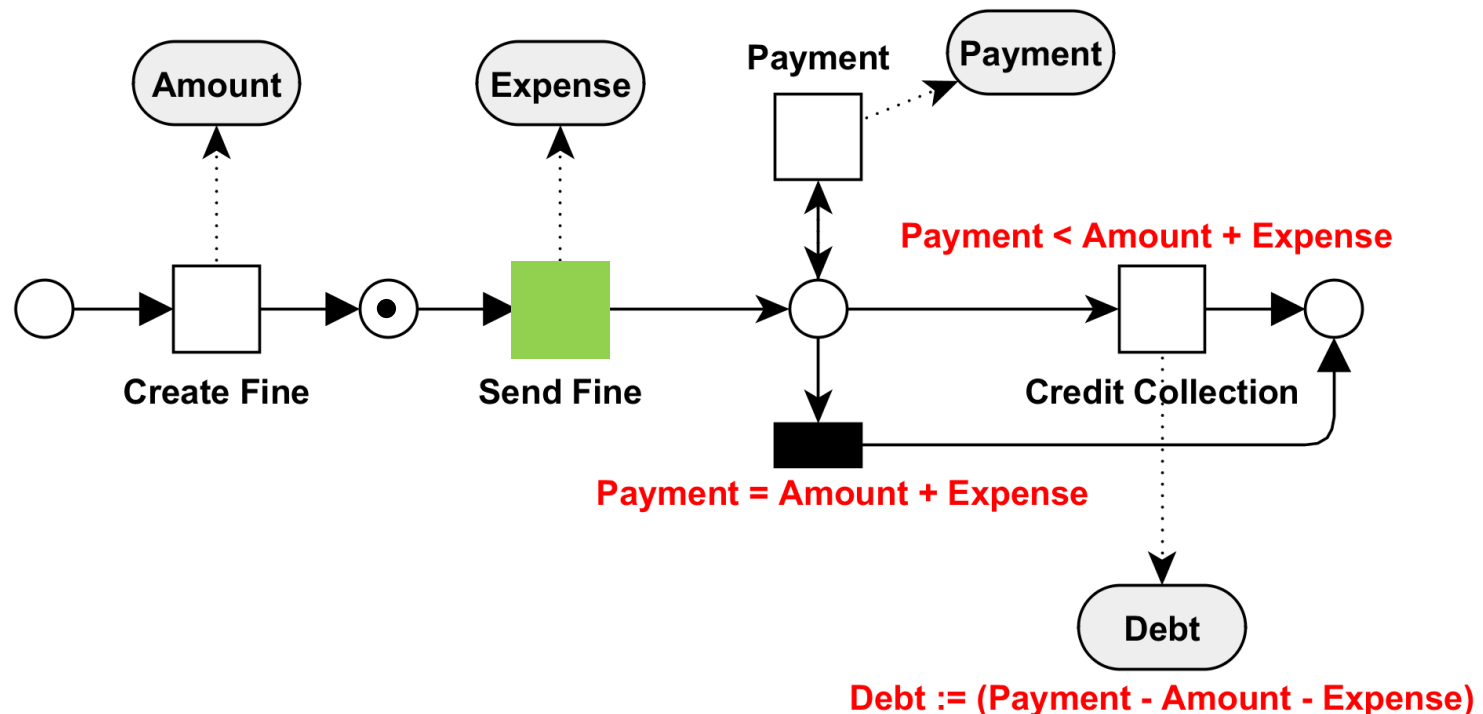
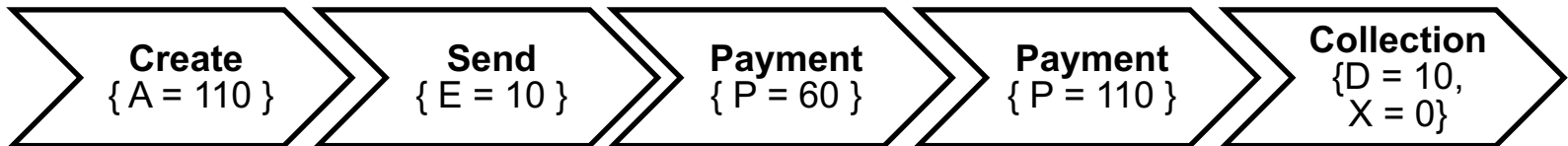


Alignment: Searching for a likely Explanation

Log

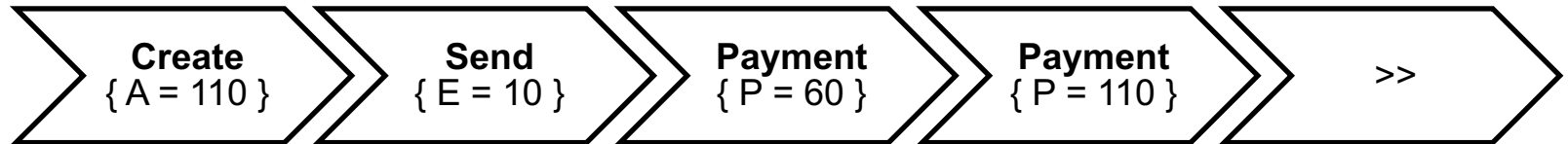


Process

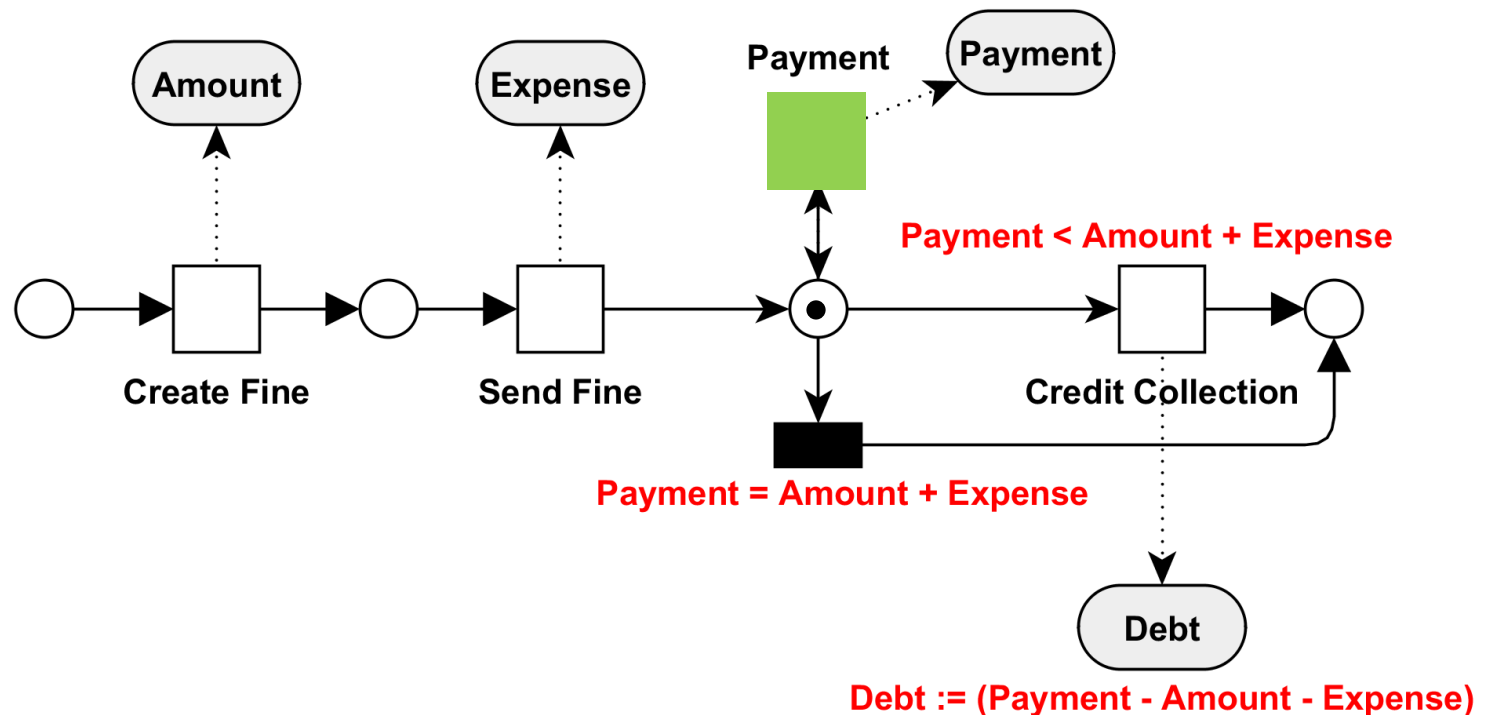
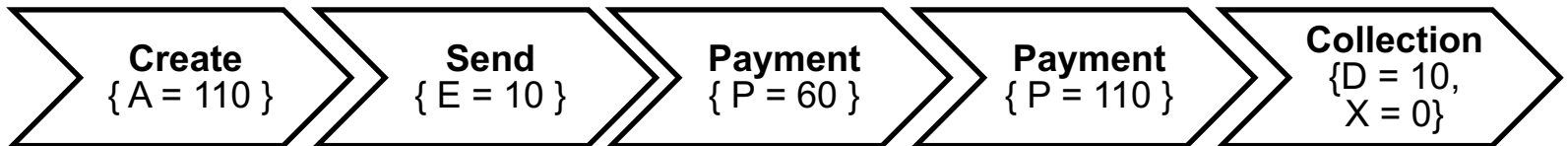


Alignment: Searching for a likely Explanation

Log

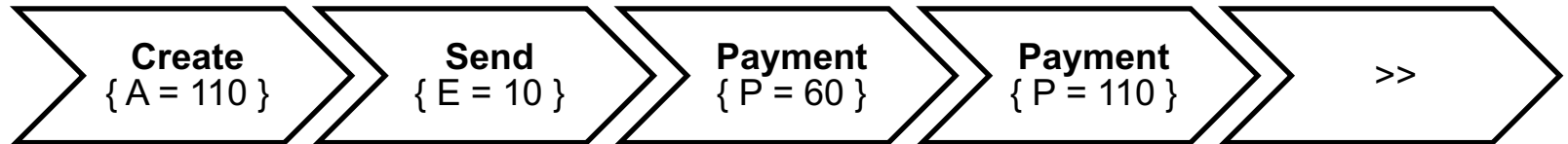


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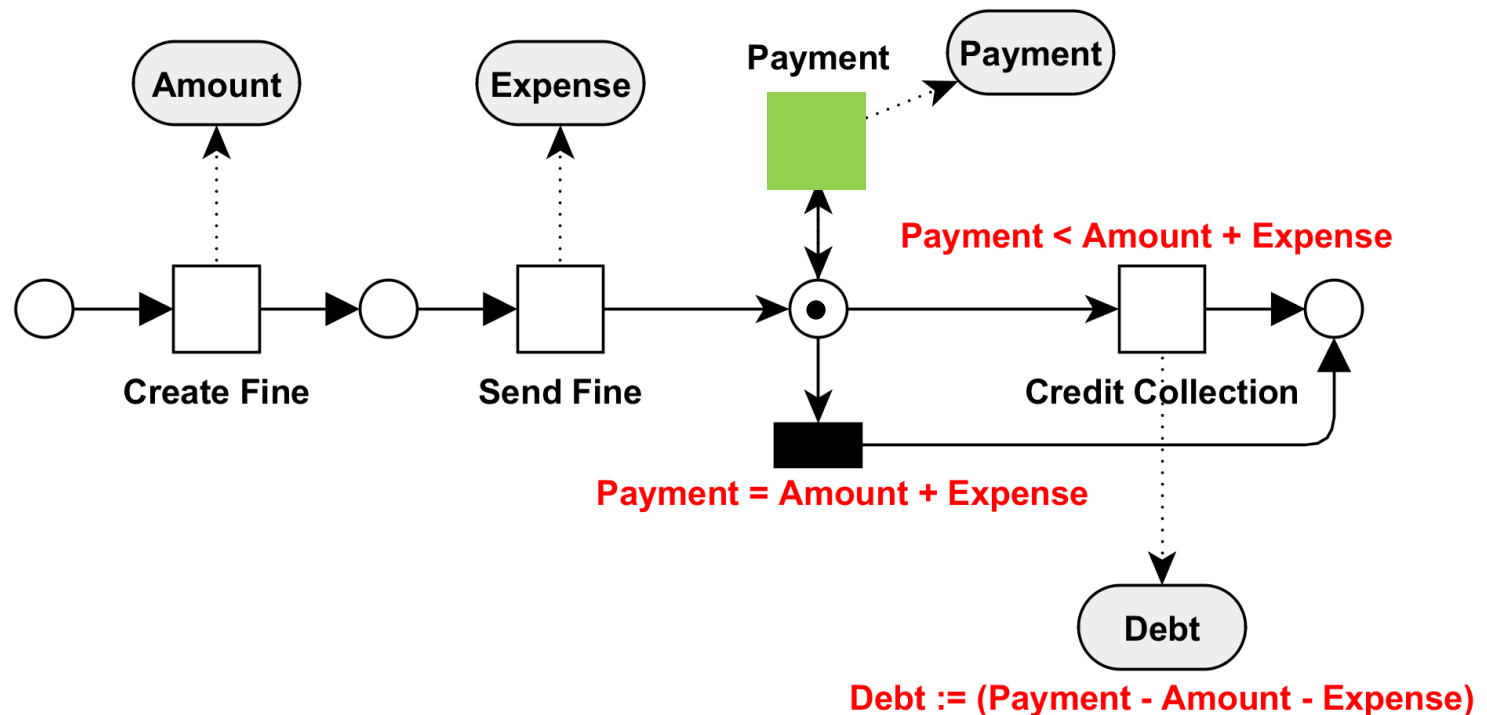
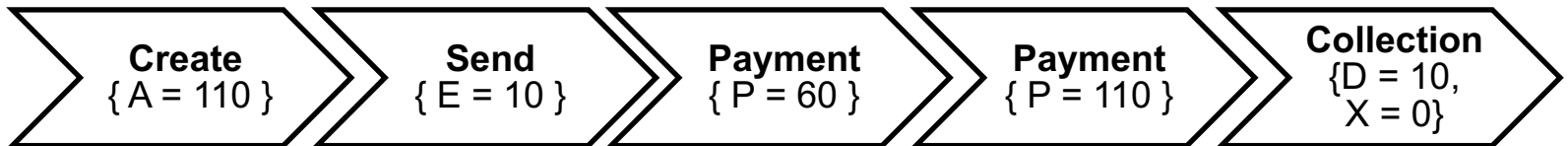


Alignment: Searching for a likely Explanation

Log

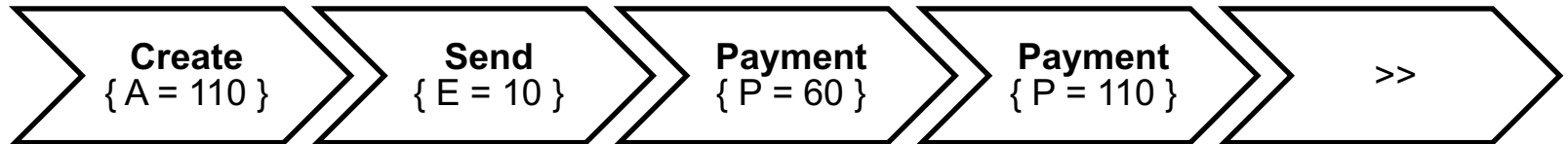


Process

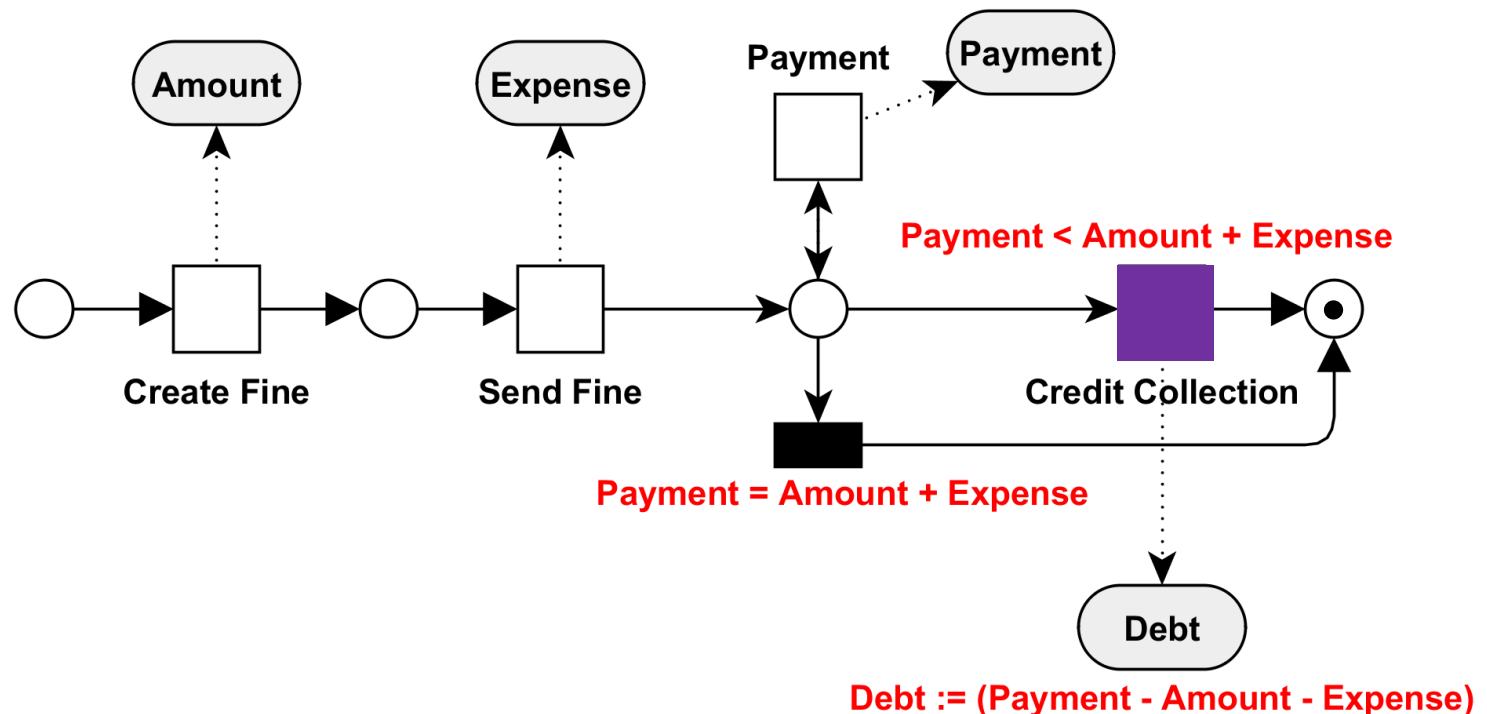
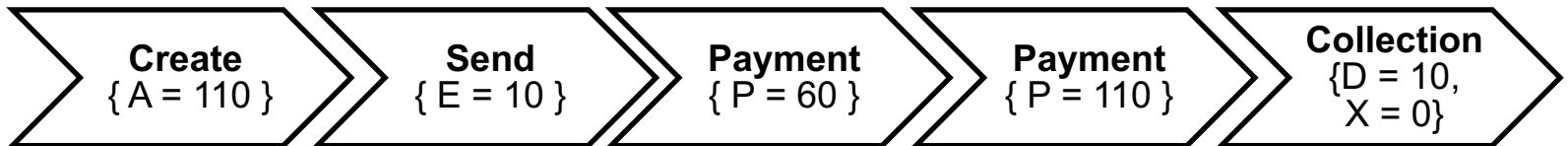


Alignment: Searching for a likely Explanation

Log

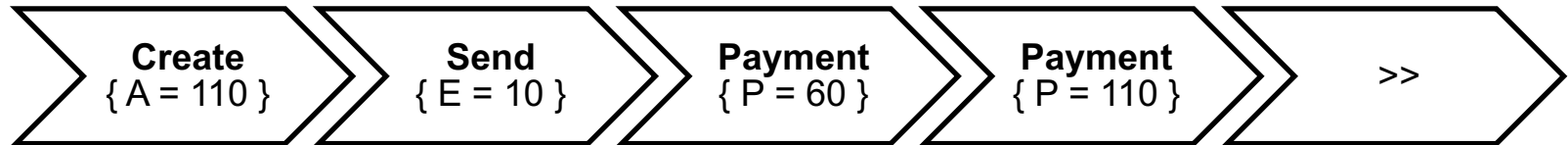


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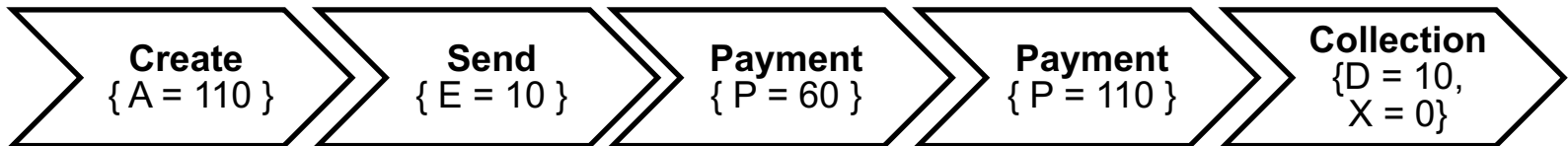


Alignment: Searching for a likely Explanation

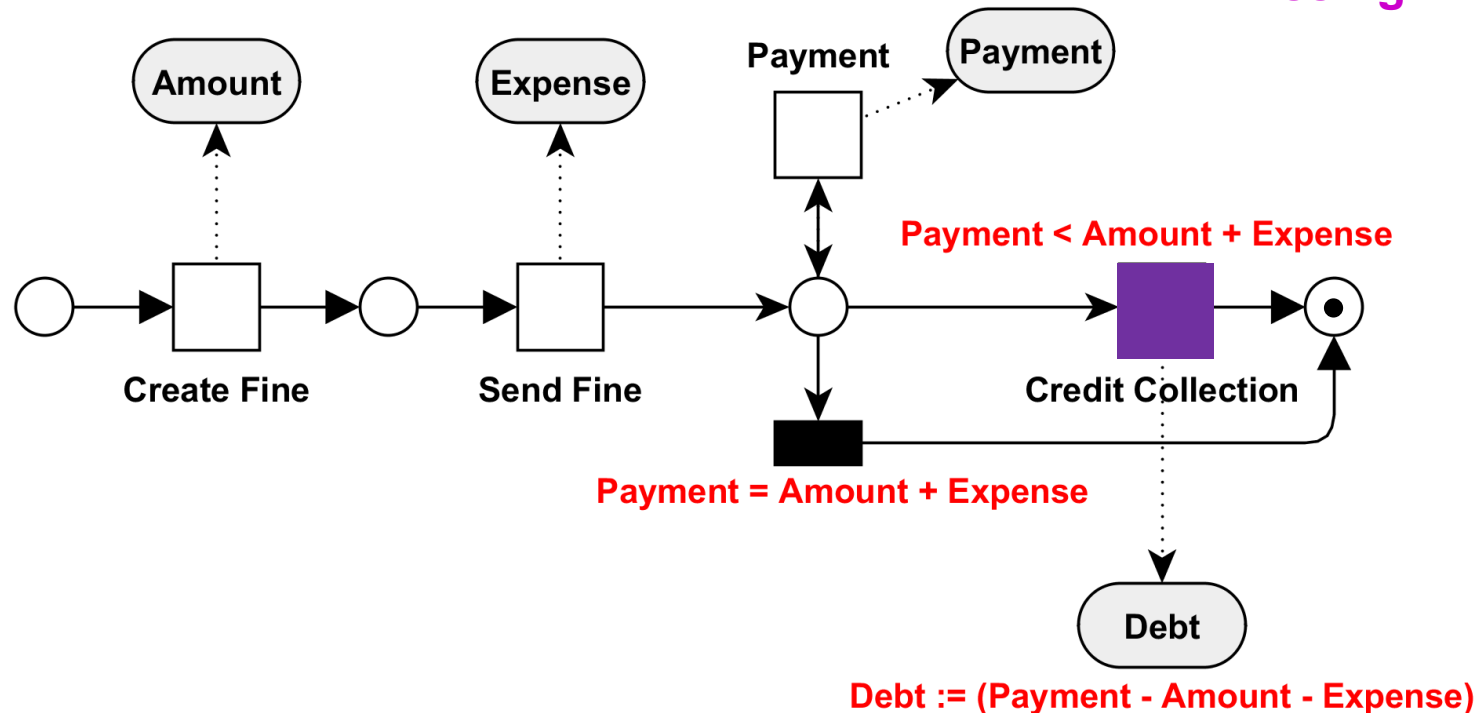
Log



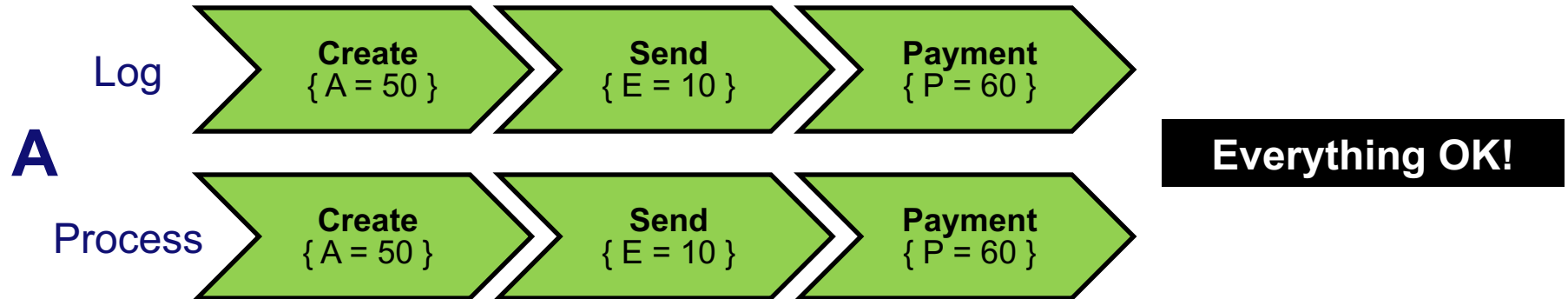
Process



Missing Event



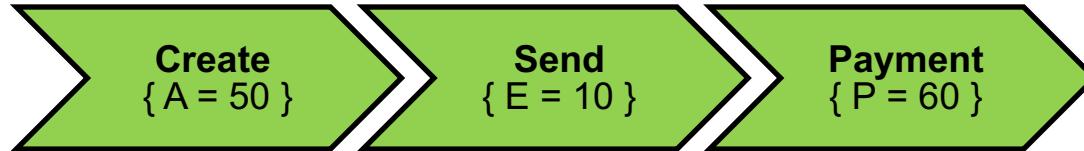
Example of an Alignment



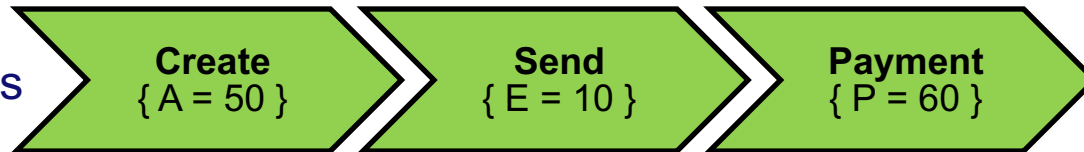
Example of an Alignment

A

Log



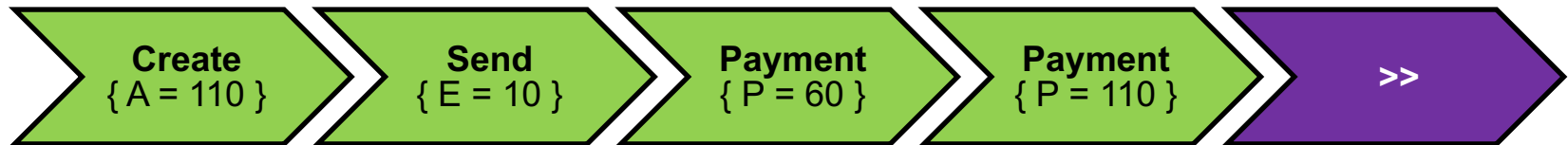
Process



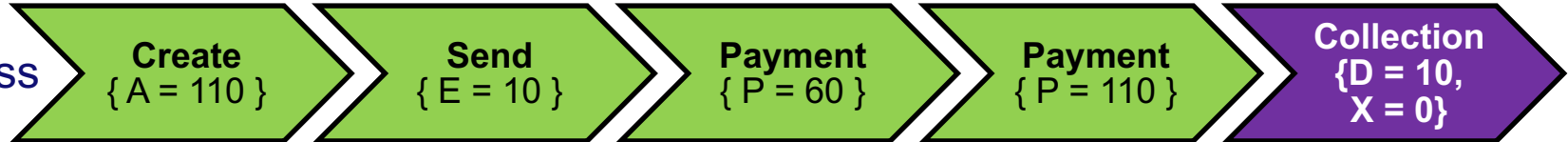
Everything OK!

B

Log

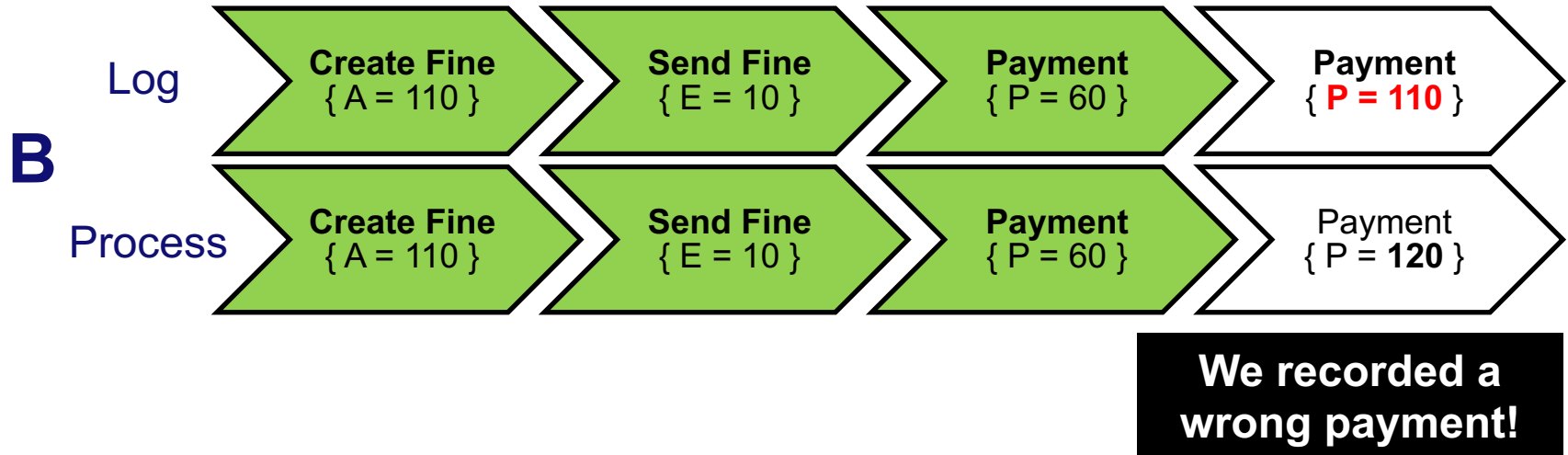


Process

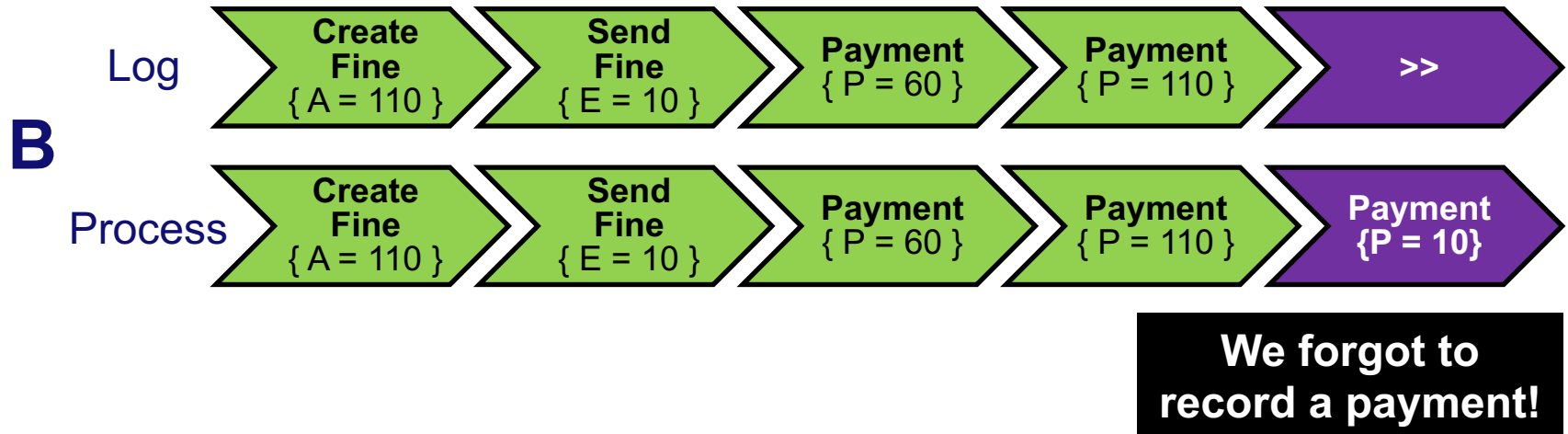
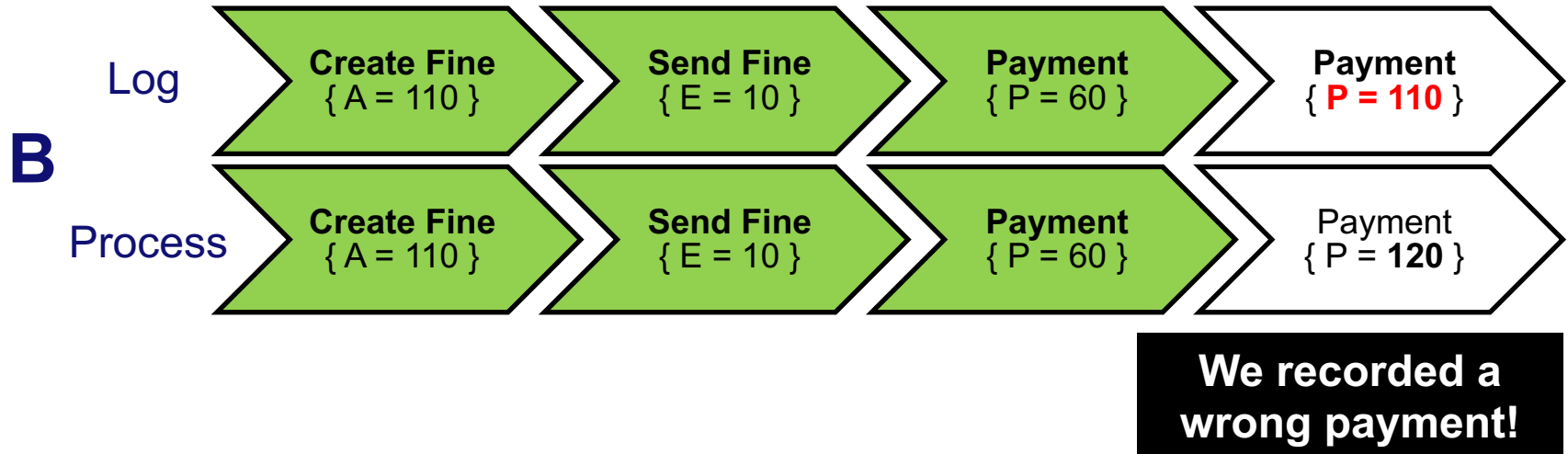


**We forgot to
collect 10 EUR!**

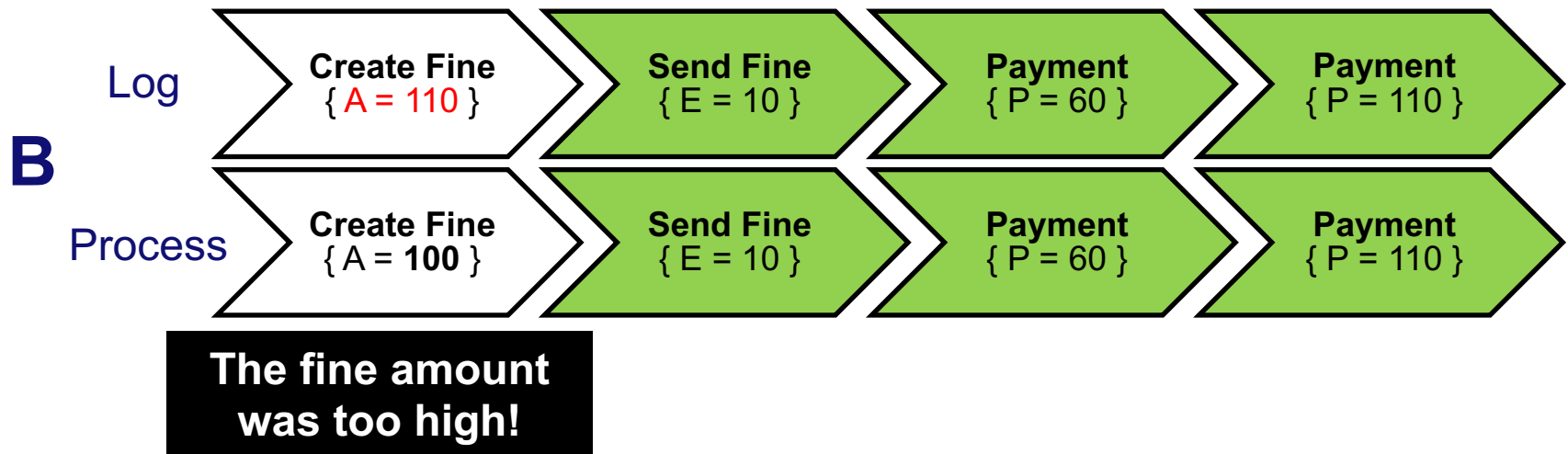
More Alignments?



More Alignments?



There is even more!



Domain Knowledge